MANASWI KIRANKUMAR KATKAR

B.E. – Artificial Intelligence & Data Science

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PROFESSIONAL SUMMARY

B.E. Artificial Intelligence & Data Science student with hands-on experience in **C, C++, Python, Java and SQL**, with a strong foundation in **Data Structures & Algorithms, OOP, DBMS, Operating Systems and Computer Networks**. Experienced in developing Java-based database applications using **JDBC, MySQL and Swing**, along with Python-based computer vision projects. Strong problem-solving skills with practical experience in software development, data analysis and team-based project environments.

EDUCATION

B.E. Artificial Intelligence & Data Science 2024 - 2027

D. Y. Patil Institute of Engineering, Management and Research, Pune | **CGPA: 8.35**

HSC 2022 – 2023

C.E.S. Prerana Jr. college | **61.33%**

SSC 2020-2021

C.E.S. Prerana High school | **95.80%**

TECHNICAL SKILLS

Programming Languages: C, C++, Python, Java
Data Structures & Algorithms: DSA, Problem Solving
Core CS: OOP, DBMS, Operating Systems, Computer Networks
Databases: MySQL, MongoDB
Java Technologies: JDBC, Swing
Web Technologies: HTML, CSS
AI & Computer Vision: Machine Learning, Deep Learning, Computer Vision, OpenCV, YOLO, CNN, TensorFlow
Tools: Git, VS Code
Systems: Linux , Embedded Systems , Microcontrollers

CERTIFICATIONS

C Programming Certification
C++ Programming Certification

EXPERIENCE

Data Analyst Intern — MEGSYS IT / Computers / Software **Jan 2026 – Feb 2026**

Worked with real-world datasets for **data analysis, preprocessing and visualization**.
Performed data cleaning and preparation to improve data quality.
Supported decision-making through data analysis and interpretation.
Developed practical understanding of working with structured datasets and analytical workflows.

PROJECTS

College Event Management System
Technologies: Java, Swing, MySQL, JDBC

- Built a Java Swing desktop application to manage students, events, and registrations using MySQL.
- Implemented CRUD operations, JDBC connectivity, `PreparedStatement`, input validation, and relational database design using primary and foreign keys.
- Designed an interactive dashboard and GUI for efficient event and registration management.

AI-Based Crop & Weed Detection for Autonomous Farming Machine
Technologies: Python, OpenCV, YOLO, Computer Vision

- Developed an AI-based crop and weed detection system for an **autonomous precision farming machine**.
- Built an image-based object detection model to distinguish **onion crops from unwanted weeds**.
- Used **Python, OpenCV and YOLO** for image processing, model development and real-time detection.
- Worked as part of a multidisciplinary team integrating the AI component with the autonomous machine system.
- Contributed to the development of a practical solution for automated agricultural operations.

Smart Image Classification & Gallery Organizer
Technologies: Python, OpenCV, CNN, Machine Learning

- Developed an image classification system to automatically organize images into categories such as **selfies, objects, skies and trees**.
- Applied image processing and CNN-based classification techniques for automatic image recognition.
- Worked on model development and performance improvement for practical image categorization.

ACHIEVEMENTS

- SMART INDIA HACKATHON (SIH)- INTERNAL ROUND WINNER

WORKSHOPS AND TRAINING

Agentic AI Workshop – Progression School **July 2025**

- Learned fundamentals of **Agentic AI and autonomous AI systems**.
- Gained exposure to intelligent agents and their real-world applications.

Digital Productivity – NIIT Foundation **September 2024**

- Learned productivity techniques involving time management, task organization and digital tools.